الإمارات العربية المتحدة
وزارة التربية والتعليم

الميثاق الوطني للتعليم

FORMATIVE ASSESSMENT 2 · UNIT 2

Formative 2 Preparation

Grade 10 Advanced · Mathematics

كتيّب التجهّز للتقويم التكويني الثاني — الوحدة الثانية

1 2 3 4 5

5	20	45	15
objectives أهداف	marks علامة	minutes دقيقة	practice questions تدريبًا

Mr. Yousef Al Yousef · **Thursday 29 October 2026**

One page per objective · every solution written out step by step

Quiz map

5 objectives = 5 recipes · 5 أهداف = 5 وصفات

1 **Interior angles of a polygon** الزوايا الداخلية
Sum = (n – 2) · 180° · set the angles equal to it · solve for x

2 **Exterior angles of a polygon** الزوايا الخارجية
Sum = 360° · regular polygon: n = 360 ÷ (180° – interior)

3 **Sides and angles of parallelograms** أضلاع وزوايا متوازي الأضلاع
Opposite sides and angles equal · consecutive angles add to 180°

4 **Tests for parallelograms** اختبارات متوازي الأضلاع
The five tests · name the test · one pair parallel + the other congruent is NOT a test

5 **Properties of kites** الطائرة الورقية
The angles between unequal sides are equal · the four angles add to 360°

1 · Learn
Read the idea and the recipe
اقرأ الفكرة والوصفة

2 · Watch
Cover the solution, try it, then compare
غطِّ الحل وحاول ثم قارن

3 · Try
Do “Your turn”, check at the back
حلّ التدريبات وصحّح

1 Interior angles of a polygon

الزوايا الداخلية لمضلع

The interior angles of an n-sided polygon add up to (n – 2) · 180°. Write the equation, then solve for x.

Sum of the interior angles = (n – 2) · 180°. Add the angle expressions, put them equal to that sum, solve for x, then substitute to find each angle.

مجموع الزوايا الداخلية = (n – 2) · 180°. اجمع تعابير الزوايا وساوِها بالمجموع، حلّ لإيجاد x ثم عوّض.

Recipe الوصفة

1. **Count the sides n and find (n – 2) · 180**
   عُدّ الأضلاع n واحسب (n – 2) · 180
2. **Add all the angle expressions and set the sum equal**
   اجمع التعابير وساوِ بالمجموع
3. **Solve the equation for x**
   حلّ المعادلة لإيجاد x
4. **Substitute x to get each angle, then check that they add to the sum**
   عوّض لإيجاد كل زاوية ثم تحقّق من المجموع

Watch out

- Write a right angle as 90°, not as x.
- Check at the end: the angles you found must add up to the sum.

Tool. S = (n – 2) · 180°

Worked example Authored (shape of Ex 1-4)

The angles of quadrilateral $ABCD$ measure $x°$, $2x°$, $(3x - 20)°$ and $(4x - 40)°$. Find the measure of each interior angle.

1. a quadrilateral has 4 sides
   $S = (4 - 2) \cdot 180 = 360°$
2. the four angles add to 360°
   $x + 2x + (3x - 20) + (4x - 40) = 360$
3. add 60, divide by 10
   $10x - 60 = 360 \Rightarrow x = 42$
4. substitute x = 42 into each expression
   $42°, 84°, 106°, 128°$
5. check the sum
   $42 + 84 + 106 + 128 = 360$ ✓

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 1) Find the sum of the interior angles of a decagon (10 sides).
2. Authored (shape of Ex 3) A pentagon has angles $x°$, $(x + 20)°$, $(2x - 10)°$, $100°$ and $110°$. Find x and the measure of each angle.
3. Authored (shape of Ex 2) Quadrilateral $JKLM$ has angles $(4x + 5)°$, $(2x + 15)°$, $(3x - 10)°$ and $x°$. Find the measure of the largest angle.

2 Exterior angles of a polygon
الزوايا الخارجية لمضلع

The exterior angles of ANY polygon add up to 360°. In a regular polygon every exterior angle is 360 ÷ n.

An exterior angle and its interior angle add to 180°. The exterior angles (one at each vertex) always add to 360°. Regular polygon: number of sides n = 360 ÷ (exterior angle).

الزاوية الخارجية والداخلية مجموعهما 180°، ومجموع الزوايا الخارجية لأي مضلع 360°. مضلع منتظم: n = 360 ÷ الزاوية الخارجية.

Recipe الوصفة

1. **Interior angle given: exterior = 180° − interior**
   الخارجية = 180° − الداخلية
2. **Regular polygon: n = 360 ÷ exterior angle**
   مضلع منتظم: n = 360 ÷ الخارجية
3. **Exterior angles in a diagram: add them and set the sum equal to 360**
   اجمع الزوايا الخارجية وساوِ بـ 360
4. **Solve for x; check that they add to 360°**
   حلّ لإيجاد x ثم تحقّق

Watch out

- Use 360°, not (n − 2) · 180°, when the angles are exterior.
- Only a REGULAR polygon has equal exterior angles.

Tool. sum of exterior angles = 360° · n = 360 ÷ exterior

Worked example Authored (shape of Ex 9-18)

The measure of an interior angle of a regular polygon is $140°$. Find the number of sides in the polygon.

1. interior + exterior = 180°

$$\text{exterior angle} = 180 - 140 = 40°$$

2. the exterior angles add to 360° and are all equal

$$n = \frac{360}{40}$$

3. a regular 9-gon

$$n = 9 \text{ sides}$$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 11) The measure of an interior angle of a regular polygon is $165°$. Find the number of sides.
2. Authored (shape of Ex 10) The measure of an interior angle of a regular polygon is $120°$. Find the number of sides.
3. Authored (shape of Ex 15) A pentagon has exterior angles $x°$, $2x°$, $(3x - 10)°$, $(x + 30)°$ and $(2x - 20)°$. Find x.

3 Sides and angles of parallelograms

أضلاع وزوايا متوازي الأضلاع

In a parallelogram opposite sides are congruent, opposite angles are congruent, consecutive angles add to 180°.

Opposite sides are equal; opposite angles are equal; consecutive angles are supplementary; the diagonals bisect each other.

الأضلاع المتقابلة متساوية، والزوايا المتقابلة متساوية، والمتتالية مجموعها 180°، والقطران ينصّف كلٌّ منهما الآخر.

Recipe الوصفة

1. **Mark the opposite sides and the opposite angles**
   علّم الأضلاع والزوايا المتقابلة
2. **Opposite sides (or angles): set the two expressions equal**
   ساوِ تعبيري الضلعين المتقابلين
3. **Consecutive angles: set their sum equal to 180°**
   المتتالية: مجموعها 180°
4. **Solve each variable separately**
   حلّ كل متغيّر على حدة

Watch out

- Equal expressions are OPPOSITE sides, not neighbouring ones.
- Solve a and b with their own equations.

Tool. opposite sides: = · consecutive angles: sum = 180°

Worked example Authored (shape of Ex 9-12)

In parallelogram $ABCD$, $AB = 2a + 3$, $CD = a + 11$, $BC = 4b - 1$ and $AD = 3b + 5$. Find a and b.

1. opposite sides are congruent
   $AB = CD:\ 2a + 3 = a + 11$
2. subtract a, subtract 3
   $a = 8$
3. the other pair of opposite sides
   $BC = AD:\ 4b - 1 = 3b + 5$
4. subtract $3b$, add 1
   $b = 6$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 11) In parallelogram $ABCD$, $m\angle A = (3x + 15)°$ and $m\angle B = (2x + 25)°$. Find x.
2. Authored (shape of Ex 11) In parallelogram $FGHD$, $m\angle F = (4x - 10)°$ and $m\angle H = (2x + 30)°$. Find x.
3. Authored (shape of Ex 12) The diagonals of parallelogram $JKLM$ meet at E, with $JE = 3y + 1$ and $EL = 5y - 7$. Find y.

4 Tests for parallelograms

اختبارات متوازي الأضلاع

Lesson 2-3 · Part 1 · pp. 75–79
MAT.3.09.01.012

Five tests make a quadrilateral a parallelogram. Name the test you used, and check that the marks really give it.

A quadrilateral is a parallelogram if: both pairs of opposite sides are parallel (definition) · both pairs of opposite sides are congruent · both pairs of opposite angles are congruent · the diagonals bisect each other · one pair of opposite sides is BOTH parallel and congruent.

الشكل الرباعي متوازي أضلاع إذا: ضلعاه المتقابلان متوازيان (تعريف) أو متطابقان في الزوجين، أو الزاويتان المتقابلتان متطابقتان في الزوجين، أو القطران ينصّف كلٌّ منهما الآخر، أو ضلعان متقابلان متوازيان ومتطابقان.

Recipe الوصفة

1. **Read the marks: ticks = congruent, arrows = parallel**
   اقرأ العلامات: الشُّرَط تطابق، الأسهم توازي
2. **Match the marks to one of the five tests**
   طابقها مع أحد الاختبارات الخمسة
3. **Both pairs (or one pair that is parallel AND congruent) → yes, name the test**
   زوجان كاملان → نعم واذكر الاختبار
4. **Only one pair, or one condition of each kind on different pairs → no**
   زوج واحد فقط → لا

Watch out

- One pair parallel and the OTHER pair congruent is NOT a test (it can be an isosceles trapezoid).
- Opposite means facing each other, not neighbouring.

Tool. the five tests · Theorems 2.9 – 2.12

Worked example Authored (shape of Ex 1-6)

In quadrilateral $WXYZ$, $\overline{WX} \parallel \overline{ZY}$ and $WX = ZY$. Is it a parallelogram? Justify your answer.

1. one pair of opposite sides is parallel
   $\overline{WX} \parallel \overline{ZY}$ (given)
2. the same pair is congruent
   $WX = ZY$ (given)
3. both conditions on the SAME pair
   one pair of opposite sides is parallel and congruent
4. yes: name the theorem
   so $WXYZ$ is a parallelogram (Theorem 2.12)

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 5) The diagonals of quadrilateral $ABCD$ meet at E with $AE = EC$ and $BE = ED$. Is it a parallelogram? Justify.
2. Authored (shape of Ex 6) In quadrilateral $PQRS$, $\overline{PQ} \parallel \overline{SR}$ and $QR = PS$. Is it a parallelogram? Justify.
3. Authored (shape of Ex 3) In quadrilateral $EFGH$, only $EF = GH$ is known. Is it a parallelogram? Justify.

5 Properties of kites

خصائص الطائرة الورقية

Lesson 2-6 · Part 3 · pp. 105–106
MAT.3.09.01.019

A kite has two pairs of congruent ADJACENT sides. Exactly one pair of opposite angles is congruent, and the four angles add to 360°.

In kite ABCD with AB = AD and CB = CD, the congruent angles are B and D (between the unequal sides). A + C + 2·B = 360°.

في الطائرة الورقية ABCD حيث AB = AD وCB = CD: الزاويتان B وD متطابقتان، والزوايا الأربع مجموعها 360°.

Recipe الوصفة

1. **Mark the equal sides: AB = AD and CB = CD**
   علّم الأضلاع المتساوية
2. **The congruent angles are the ones between UNEQUAL sides (B and D)**
   المتطابقتان بين الضلعين غير المتساويين
3. **Write A + C + B + D = 360° and use B = D**
   اكتب المجموع 360° واستخدم B = D
4. **Solve for the missing angle; check the sum**
   حلّ ثم تحقّق من المجموع

Watch out

- The angles at the ends of the line of symmetry (A and C) are NOT the equal ones.
- The answer for the pair is half of what is left, not all of it.

Tool. m∠A + m∠B + m∠C + m∠D = 360°

Worked example Authored (shape of Ex 13-16)

Kite $ABCD$ has $AB = AD$ and $CB = CD$, with $m\angle A = 50°$ and $m\angle C = 100°$. Find $m\angle B$ and $m\angle D$.

1. the angles between the unequal sides are congruent
   $m\angle B = m\angle D$
2. the four angles add to 360°
   $50 + 100 + 2\,m\angle B = 360$
3. subtract 150
   $2\,m\angle B = 210$
4. divide by 2
   $m\angle B = m\angle D = 105°$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 13) Kite $PQRS$: $PQ = PS$, $RQ = RS$, $m\angle P = 120°$ and $m\angle R = 70°$. Find $m\angle Q$.
2. Authored (shape of Ex 14) Kite $ABCD$: $AB = AD$, $CB = CD$, $m\angle A = 44°$ and $m\angle C = 90°$. Find $m\angle D$.
3. Authored (shape of Ex 16) Kite $ABCD$: $AB = AD$, $CB = CD$, $m\angle A = 80°$ and $m\angle B = 108°$. Find $m\angle C$.

Solutions

written out step by step · الحلول

Objective 1 · Interior angles of a polygon — الزوايا الداخلية لمضلع

1.1 Authored (shape of Ex 1)

Find the sum of the interior angles of a decagon (10 sides).

1. $n - 2 = 8$

$S = (10 - 2) \cdot 180 = 8 \cdot 180 = 1440^\circ$

1.2 Authored (shape of Ex 3)

A pentagon has angles x°, $(x + 20)^\circ$, $(2x - 10)^\circ$, 100° and 110°. Find x and the measure of each angle.

1. a pentagon: 540°

$S = (5 - 2) \cdot 180 = 540^\circ$

2. subtract 220, divide by 4

$x + (x + 20) + (2x - 10) + 100 + 110 = 540 \Rightarrow 4x + 220 = 540 \Rightarrow x = 80$

3. the angles add to 540°

$80^\circ,\ 100^\circ,\ 150^\circ,\ 100^\circ,\ 110^\circ$

1.3 Authored (shape of Ex 2)

Quadrilateral $JKLM$ has angles $(4x + 5)^\circ$, $(2x + 15)^\circ$, $(3x - 10)^\circ$ and x°. Find the measure of the largest angle.

1. the four angles add to 360°

$(4x + 5) + (2x + 15) + (3x - 10) + x = 360$

2. subtract 10, divide by 10

$10x + 10 = 360 \Rightarrow x = 35$

3. the angles are $145^\circ, 85^\circ, 95^\circ, 35^\circ$

$4(35) + 5 = 145^\circ$ is the largest

Objective 2 · Exterior angles of a polygon — الزوايا الخارجية لمضلع

2.1 Authored (shape of Ex 11)

The measure of an interior angle of a regular polygon is 165°. Find the number of sides.

1. interior + exterior = 180°

exterior $= 180 - 165 = 15^\circ$

2. the exterior angles add to 360°

$n = \dfrac{360}{15} = 24$

2.2 Authored (shape of Ex 10)

The measure of an interior angle of a regular polygon is 120°. Find the number of sides.

1. interior + exterior = 180°

   $\text{exterior} = 180 - 120 = 60^\circ$

2. a regular hexagon

   $n = \dfrac{360}{60} = 6$

2.3 Authored (shape of Ex 15)

A pentagon has exterior angles x°, $2x^\circ$, $(3x-10)^\circ$, $(x+30)^\circ$ and $(2x-20)^\circ$. Find x.

1. the exterior angles add to 360°

   $x + 2x + (3x-10) + (x+30) + (2x-20) = 360$

2. the angles are $40^\circ, 80^\circ, 110^\circ, 70^\circ, 60^\circ$

   $9x = 360 \Rightarrow x = 40$

Objective 3 · Sides and angles of parallelograms

أضلاع وزوايا متوازي الأضلاع

3.1 Authored (shape of Ex 11)

In parallelogram $ABCD$, $m\angle A = (3x+15)^\circ$ and $m\angle B = (2x+25)^\circ$. Find x.

1. consecutive angles add to 180°

   $\angle A$ and $\angle B$ are consecutive: $(3x+15) + (2x+25) = 180$

2. subtract 40, divide by 5

   $5x + 40 = 180 \Rightarrow x = 28$

3.2 Authored (shape of Ex 11)

In parallelogram $FGHD$, $m\angle F = (4x-10)^\circ$ and $m\angle H = (2x+30)^\circ$. Find x.

1. opposite angles are congruent

   $\angle F \cong \angle H:\ 4x - 10 = 2x + 30$

2. subtract 2x, add 10, divide by 2

   $2x = 40 \Rightarrow x = 20$

3.3 Authored (shape of Ex 12)

The diagonals of parallelogram $JKLM$ meet at E, with $JE = 3y + 1$ and $EL = 5y - 7$. Find y.

1. the diagonals bisect each other

   $JE = EL:\ 3y + 1 = 5y - 7$

2. subtract 3y, add 7, divide by 2

   $8 = 2y \Rightarrow y = 4$

Objective 4 · Tests for parallelograms — اختبارات متوازي الأضلاع

4.1 Authored (shape of Ex 5)

The diagonals of quadrilateral $ABCD$ meet at E with $AE = EC$ and $BE = ED$. Is it a parallelogram? Justify.

1. each diagonal is cut in two equal parts
   $AE = EC$ and $BE = ED$
2. yes: name the theorem
   the diagonals bisect each other ⇒ a parallelogram (Theorem 2.11)

4.2 Authored (shape of Ex 6)

In quadrilateral $PQRS$, $\overline{PQ} \parallel \overline{SR}$ and $QR = PS$. Is it a parallelogram? Justify.

1. one pair parallel
   $\overline{PQ} \parallel \overline{SR}$: one pair of opposite sides is parallel
2. the conditions are on different pairs
   $QR = PS$: the OTHER pair is congruent
3. it could be an isosceles trapezoid
   not one of the tests: not necessarily a parallelogram

4.3 Authored (shape of Ex 3)

In quadrilateral $EFGH$, only $EF = GH$ is known. Is it a parallelogram? Justify.

1. one pair of equal sides only
   $EF = GH$: only one pair of opposite sides is congruent
2. a test needs both pairs, or one pair parallel and congruent
   this is not enough: not necessarily a parallelogram

Objective 5 · Properties of kites — خصائص الطائرة الورقية

5.1 Authored (shape of Ex 13)

Kite $PQRS$: $PQ = PS$, $RQ = RS$, $m\angle P = 120°$ and $m\angle R = 70°$. Find $m\angle Q$.

1. the angles between unequal sides are congruent
   $m\angle Q = m\angle S$
2. the four angles add to $360°$
   $120 + 70 + 2\,m\angle Q = 360 \Rightarrow m\angle Q = 85°$

5.2 Authored (shape of Ex 14)

Kite $ABCD$: $AB = AD$, $CB = CD$, $m\angle A = 44°$ and $m\angle C = 90°$. Find $m\angle D$.

1. B and D are the congruent pair

   $m\angle D = m\angle B$

2. the four angles add to 360°

   $44 + 90 + 2\,m\angle D = 360 \Rightarrow m\angle D = 113°$

5.3 Authored (shape of Ex 16)

Kite $ABCD$: $AB = AD$, $CB = CD$, $m\angle A = 80°$ and $m\angle B = 108°$. Find $m\angle C$.

1. B and D are congruent

   $m\angle D = m\angle B = 108°$

2. the four angles add to 360°

   $80 + 108 + 108 + m\angle C = 360 \Rightarrow m\angle C = 64°$

Ready?

Tick each line · ضع علامة

- [ ] **I can use the angle sum of a polygon to find x**
  أستطيع استخدام مجموع زوايا المضلع لإيجاد x
- [ ] **I can use the exterior angles of a polygon**
  أستطيع استخدام الزوايا الخارجية للمضلع
- [ ] **I can use the sides and angles of a parallelogram**
  أستطيع استخدام أضلاع وزوايا متوازي الأضلاع
- [ ] **I can decide, with a reason, whether a quadrilateral is a parallelogram**
  أستطيع تحديد ما إذا كان الشكل متوازي أضلاع مع التعليل
- [ ] **I can find the angles of a kite**
  أستطيع إيجاد زوايا الطائرة الورقية

On the day

Four questions are on the computer and one is on paper. Read each figure carefully: the marks on it (ticks, arrows, right angles) tell you what is given.

Every question

Name the property or the test you use. On paper, write every line: the first step already earns a mark.

One mark for every written step

لكل خطوة تكتبها علامة